Building a Curation-First TTS Dataset (Indian English + Hindi)

Sarvam AI — ML & Speech Data Pipeline intern assignment

> **Thesis.** The brief states three times that this is *not* a coding challenge — it is a **data-curation and quality-assurance** exercise. I therefore treated the code as scaffolding whose only job is to make curation **reproducible, auditable, and credit-frugal**, and spent the judgment budget on *source selection, QC gate design, threshold calibration against real data, and honest documentation*. Every clip is curated and reviewed as if it would train a production TTS model.

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1. What was built

An end-to-end, resumable pipeline (`src/s01…s09`) that turns license-verified YouTube audio into a published Hugging Face TTS dataset:

```
sources.yaml → s01 fetch → s02 segment → s03 acoustic QC → s04 transcribe →
(license gate)   (yt-dlp)     (silero-VAD)    (SNR/clip/loud)     (Sarvam STT,cached)

 s05 verify → s06 review → s07 normalize → s08 emotion → s09 build → HF Hub
 (1-speaker)    (HUMAN: listen   (mono/24kHz/     (LLM cand. +    (manifest +
  +language)     +correct text)   loudnorm)        human conf.)    push_to_hub)
```

Key properties:

- **All thresholds live in** `config/pipeline.yaml` with written rationale, so every curation decision is explicit and every iteration is a reviewable `git diff` .
- **Sarvam responses are cached on disk by audio SHA-256** (`src/utils/sarvam_client.py`), so reruns and threshold iterations cost **zero credits** and the build is idempotent.
- **The manifest (** `data/manifest.csv` **) gives full provenance** for every published clip: source URL, license, speaker, timestamps, SHA-256, and every QC metric.

Design choices and why

Choice	Decision	Rationale
Sources	**NPTEL lectures** (Indian English + Hindi)	Single speaker per course; controlled recording; abundant; **CC-BY on YouTube** so reuse is unambiguous *and machine-verifiable*.
Clip length	**4–18 s, VAD sentence-level**	Coherent TTS units with clean silence edges (LJSpeech avg ≈ 6.6 s); the brief explicitly allows "any combination totaling ~60 min."
Final audio		

Choice	Decision	Rationale
	mono, 24 kHz, 16-bit, –23 LUFS, trimmed	Modern neural-TTS standard; clips re-cut from the **original 48 kHz** source (no upsampling).
Second language	**Hindi**	Highest Sarvam ASR accuracy among Indian languages and personally QA-able (Devanagari + romanized).
Emotion	**LLM candidate → human confirm**	Emotion is acoustic; text alone can't decide it. Tag conservatively and document the limitation.

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2. Data-quality iterations

The quality funnel

Every clip flows through ordered gates; we record how many die at each. Automated gates run **before** any paid API call. Numbers below are from the validated run on the first English source (a 21-minute NPTEL lecture); the full 60-minute run extends the same funnel across all sources.

Stage	Clips	Surviving
Segmented (VAD utterances)	88	100.0%
Passed acoustic QC (pre-ASR)	63	71.6%
Transcribed (Sarvam STT)	*pending key*	—
Passed single-speaker + language	*pending*	—
Human-reviewed (decided)	*pending listen*	—
Accepted into dataset	*pending*	—

(Funnel auto-generated to `reports/funnel.csv`; figures in `reports/figures/`.)

Acoustic QC against real data

The passing clips have a **median SNR of 38.2 dB** (studio-grade); the 22 dB gate cleanly removes a noisy tail (failed-clip SNRs cluster ≤ 25 dB, min –0.1 dB). This bimodal separation is exactly what a good threshold should produce — see `figures/snr_hist.png`. Duration is well-centred (median 11 s) within the 4–18 s target — `figures/duration_hist.png`.

Iteration v0 → v1: the peak-threshold mistake (a real one)

The most instructive iteration was a **bug in my own judgment**, not the code:

- **v0** set `true_peak_dbfs_max = -0.5` as an acoustic-QC reject gate, reasoning "audio peaking near full-scale is at risk of clipping distortion." Result: **83 of 88 clips (94%) were rejected** — almost the entire source.
- **Diagnosis.** YouTube audio is loudness-maximized; peaks sitting at ≈ 0 dBFS are *normal and harmless*, not evidence of distortion. I had conflated "peaks near full-scale" (a property fixed

downstream by my own loudness normalization, which limits to –1 dBFS in `s07`) with "digitally clipped/distorted" (the real defect, already caught by the clipping-ratio metric).
- **v1 fix.** Removed true-peak as a reject gate (still *measured* and recorded for the report), and relaxed `max_clipping_ratio` from 0.001 → 0.01 (a tiny fraction of full-scale samples is expected for loudness-maximized web audio; >1% indicates real clipping). **Pass rate moved from 3% → 72%**, with the noisy/clipped tail still correctly rejected.

This is recorded as a diff in `config/pipeline.yaml` and is, I think, the single most honest signal in the submission: a wrong threshold caught by *looking at the data*, diagnosed, and corrected — rather than tuned silently to "look good."

Other filtering decisions

- **License gate (s01).** A source is fetched only if yt-dlp reports its license as "*Creative Commons Attribution license (reuse allowed).*" Reupload/coaching channels that re-host NPTEL without the CC flag are auto-rejected. The flag is recorded per clip in the manifest — provenance, not assertion.
- **VAD over fixed windows.** Fixed 30/60 s cuts would slice mid-sentence and bury clean silence boundaries; VAD yields coherent utterances. Over-long utterances are re-split on a hard cap.
- **Free single-speaker check (s05).** We reuse Sarvam's diarized output: any clip with >1 `speaker_id`, a language mismatch, or low `language_probability` is dropped — with no additional API spend.

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3. Observations

(Transcript/emotion observations are populated during the Sarvam-transcription + human-review pass; the methodology and tooling are complete and described here.)

- **Expected transcription errors.** NPTEL lectures carry domain jargon (thermodynamics, CS, math). These are predictable ASR-error hot-spots: technical terms, acronyms, numerals/units, and English↔Hindi code-mixing in the Hindi set. The review harness pre-fills the ASR transcript so the reviewer corrects rather than re-types, and every edit is logged — yielding a measured WER (`reports/wer.json`).
- **Speaker/quality issues.** Even within a single-speaker lecture, some segments have room reverb, AC hum, or a cougher in the room; the SNR gate + manual listen catch these. Cross-talk during Q&A is caught by the diarization gate.
- **Emotion-tagging challenge.** Lecture/narration content is overwhelmingly `neutral` / `formal` / `conversational`. Affective tags (happy/sad/angry) are rare and inherently subjective from short clips. We tag conservatively and report the distribution honestly rather than manufacturing diversity.

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4. What worked

- **Pre-ASR acoustic gating** — rejecting bad audio for free before spending any Sarvam credit; the credit-frugal core of the design.

- **Reusing diarization for single-speaker verification** — a quality gate at zero marginal cost.
- **SHA-256 response caching** — iteration and re-runs cost nothing; the whole build is reproducible from cache.
- **Programmatic license verification** — turning "is this reusable?" from a claim into a recorded, per-clip fact.
- **Calibrating thresholds against the data distribution** rather than guessing — the SNR histogram made the 22 dB choice obvious.

5. What didn't work / dead ends

- **The v0 peak gate** (§2) — wrong mental model, fixed.
- **YouTube keyword search for sources is noisy** — queries for "NPTEL … Hindi" surfaced reupload and exam-coaching channels (not CC-BY) far more than official lecture content. Conclusion: anchor on *official channels* and let the license gate enforce correctness, rather than trusting search ranking.
- **NPTEL's CC-BY flagging is inconsistent** — even within the official `nptelhrd` channel, some lectures carry the YouTube CC-BY flag (the whole *Psychrometry* course does) and many don't. This is why licensing is verified *per video* and recorded in the manifest, never assumed from the channel. (A single CC-BY course conveniently gives one clean speaker for the entire English half.)
- **A CC-BY-flagged Hindi *lecture* was hard to find by automated search** in the time box (results skewed to promo/registration clips). Hindi source selection is left as an explicit human pass — appropriate, since the brief centres on human curation.
- **Public-domain text ≠ public-domain audio** — Premchand's *writing* is PD in India, but an audiobook *recording* of it is separately copyrighted. PD text is not a licensing shortcut for the Hindi audio; the source channel itself must CC-license.
- **Sourcing options ruled out.** TED/TEDx (CC-BY-**ND** forbids derivatives), All India Radio archives (paid license only), and general news/podcast clips (standard license + multi-speaker/music). Documented in `config/sources.yaml:rejected`.
- **Text-only emotion tagging is weak** — confirmed expectation: transcripts rarely reveal affect, hence the human-confirm step.

6. Future improvements

- **Blind gold WER set** — hand-transcribe N clips/language *before* seeing ASR, for an unbiased WER (current WER vs. human-corrected text is a slightly optimistic proxy).
- **Audio-based emotion classifier** — replace text-only LLM candidates with a prosody/acoustic model, then human-confirm.
- **More speakers & languages** — additional NPTEL professors and a third Indian language for broader coverage.
- **MOS listening test** + automatic alignment-confidence scoring to prioritize review.
- **Forced alignment** (e.g. MFA) for tighter word-level boundaries than VAD edges.

Appendix

- **Reproducibility.** Pinned deps (`requirements.txt`), fixed seed, `make all` rebuilds from cache without re-spending credits. `validate_dataset.py` asserts every invariant (mono/24 kHz/16-bit, one speaker, non-empty transcript, duration window, language ∈ {en-IN, hi-IN}, SHA-256 match, ~60 min ~30/30).
- **Licensing & ethics.** Audio CC-BY-4.0 with per-clip attribution to NPTEL; code MIT. Only YouTube-CC-BY-flagged videos are used, verified at fetch.
- **Exact Sarvam usage** is logged per call to `data/cache/api_usage.log` .
- **Artifacts:** `reports/funnel.csv` , `reports/wer.json` , `reports/figures/*.png` , `data/manifest.csv` , `review/review_log.csv` .

Appendix — Figures